sg13g2_stdcell_hv — Generation of a Thick-Oxide (3.3 V) Standard Cell Library

IHP SG13G2 open PDK · derivation, all views, verification and physical sign-off

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1 Scope and result

The IHP SG13G2 open PDK ships no 3.3 V standard cells: its thick-oxide devices appear only in the `sg13g2_io` pad ring. `sg13g2_stdcell_hv` fills that gap — **all 84 cells** of `sg13g2_stdcell` rebuilt on `sg13_hv_nmos` / `sg13_hv_pmos`, same topology, pin names and pin order, named `sg13g2_hv_*` so both libraries coexist in one netlist.

Deliverable	Coverage	Status
SPICE netlist (`spice/`)	84 cells, 920 devices	verified against 3 independent views
CDL netlist (`cdl/`)	84 cells + 2 tie cells	LVS reference, all 84 cells match
Verilog (`verilog/`)	84 modules	shared `ihp_*` UDPs, deliberately not duplicated
xschem symbols / schematics	84 + gallery sheet	netlist-equivalence proven
GDS layout (`gds/`)	**84 cells** (66 retargeted + 18 per-cell generated)	**DRC clean, LVS clean**
LEF abstracts (`lef/`)	**84 macros** + `CoreSiteHV` site	generated from the GDS, pin sets verified against CDL
Liberty NLDM (`lib/`)	82 cells, 660 timing tables	combinational + sequential + tri-state; areas all measured
LibreLane SCL (`librelane/`)	site, cell maps, tracks, excludes	flops map natively through `dfflibmap`

Physical sign-off, one fixed invocation of the PDK's own klayout decks — on the abutted two-row array and on the stricter **shared-rail array** (rows at the true 7.14 µm pitch, mixed and mirrored vertical neighbours — what a placed block actually produces) — and, since rev. 5, under Magic's *strict analog* N-well rules as well:

Check	This library	IHP thin-oxide control (same harness)
DRC, cell rules	**0**	0
DRC, cell rules, shared-rail mixed array	**0**	—
DRC, Magic (strict analog N-well rules)	**0 `NW.*`**	—
DRC, chip-level metal density	7	6
LVS	**84** / **84 match**	fill cells fail identically without the documented relaxation

The density items are harness properties, not cell defects: metal density is a check on a *filled* die, and the one differing count (`M1.j`, min 35 % Metal1) reads 35.0 % over the padded array's bounding box against 37.1 % over the actual cell area (control: 45.2 %).

The library has also carried a real block: two variants of the `spi_slave` IP placed and routed through LibreLane/OpenROAD at 100 MHz pass the full IHP signoff deck — zero geometry violations (section 6).

2 The device transform

	thin oxide	thick oxide
device model	`sg13_lv_nmos` / `sg13_lv_pmos`	`sg13_hv_nmos` / `sg13_hv_pmos`
gate length	130 nm	**450 nm** (`Gat.a3` minimum)
NMOS width	—	unchanged
PMOS width	—	**× 2.40**, snapped per finger to the 5 nm grid
`as`/`ad`/`ps`/`pd`	—	recomputed from device geometry
supply	1.2 V	3.3 V

Deliberate long-channel devices (decap 1.0 µm, `sighold` 700 nm, `dlygate4sd3_1` 500 nm) keep their lengths; 914 devices moved to 450 nm. The antenna cell is unchanged apart from its name (junction diodes).

Why × 2.40. The thin-oxide library is sized for a centred switching threshold ($V_m/V_{DD} = 0.5046$ at $W_p/W_n = 1.5135$), not drive match. At 3.3 V / 450 nm the PMOS is far weaker (219 vs 533 µA/µm) and the same V_m needs $W_p/W_n = 3.63$, i.e. $K_p = 3.63/1.5135 = 2.40$ on every PMOS with every NMOS untouched — preserving each cell's internal stack ratios. Across the library's 0.3–17.9 µm width range the required ratio stays within ±5 % of 2.40 (`work/vm_sweep.py`, `work/ratio_check.py`).

Parasitics follow the vendor deck's geometry formulas; before any generation they were validated by recomputing all 924 thin-oxide devices (worst relative error 3.75×10^{-4}, the vendor's four printed digits). The generator refuses to run if this validation fails.

Cost, measured on `inv_1` (`work/fo4.py`): input capacitance 2.67 → 5.87 fF (2.20×), FO4 delay 53.6 → 142.4 ps (2.66×). The thick-oxide NMOS delivers *more* current per micron at 3.3 V than the thin-oxide at 1.2 V (533 vs 391 µA/µm) — the loss is capacitance, not drive.

3 The views

All netlist views are re-emitted from one internal model by `work/gen_hv_lib.py`, so a device cannot silently diverge between views.

- **`spice/`** — 84 subckts, 920 PSP103 devices (OSDI/ngspice); widths synchronised to the drawn layout.
- **`cdl/`** — the same subcircuits as `M`-cards with `*.PININFO`, the LVS reference; compared field-by-field against SPICE by `verify_sch.py`.
- **`verilog/`** — 84 modules; deliberately no copy of `sg13g2_udp.v` (the `ihp_*` UDPs are shared — a second copy would collide).
- **`sym/`, `sch/`** — 84 symbols (thin-oxide drawn geometry, thick-oxide netlist prefix, `$::SG13G2_HV_SCH` resolution) and 84 schematics plus a generated gallery sheet; three-view consistency proven on all 920 devices.
- **`gds/`** — **all 84 cells**: 66 by 1-D retarget (section 4), 18 by per-cell generators (section 4.1). Uniform 7.140 µm row height (17 tracks), every width a 0.48 µm site multiple, all contact/via cuts exactly 0.16 × 0.16 µm, every rail tap on the site-centred 0.48 µm grid (`work/fix_rail_contacts.py`, section 6).
- **`lef/`** — 84 macros + `CoreSiteHV`, generated from the GDS: pins from the pin datatypes labelled by contained text, OBS = drawn metal minus pin geometry, antenna values recomputed from the netlist (the thin-oxide numbers are wrong for 3.5× longer gates). Parsed back with klayout and pin sets verified against the CDL.
- **`lib/`** — Liberty NLDM at 3.3 V / 25 °C typical (section 7).
- **`librelane/`** — a complete LibreLane standard-cell-library configuration (section 8), installed as `libs.tech/librelane/sg13g2_stdcell_hv/`.
- **`klayout/`** — a cell-library registration macro: with the repository's `klayout/` directory on `KLAYOUT_PATH`, all 84 cells appear in KLayout's Instance dialog as library `sg13g2_stdcell_hv`.

4 Layout generation

Hand-drawn 2-D layout cannot be regenerated by placement, but it can be **1-D retargeted**: monotone piecewise-linear coordinate maps applied to every vertex preserve topology, so connectivity survives by construction (`work/layout_retarget.py`). The engineering is in the breakpoints and the exceptions:

- **x-map**: gates widen 0.13 → 0.45 µm about their centres; contacted poly pitch 0.48 → 0.80 µm. One accepted consequence: `dlygate4sd2_1` staggers two gates at overlapping x, so its PMOS lands at a legal 0.625 µm, carried consistently into every netlist view.
- **y-map**: PMOS band × 2.40 plus three channel inserts for the thick-oxide clearances (`NW.d1.dig` 0.215 µm, `pSD.j1` 0.100 µm, `pSD.i1` rail insert 0.430 µm), derived library-wide so every cell keeps one row height.
- **PMOS diffusion** is re-emitted per connected slab piece with channel slabs frozen — the only scheme that keeps exact 5 nm-grid device widths without merging, collapsing or renaming devices (each alternative was tried and caught by LVS or `Act.*` rules).
- **Cut layers are translated, never scaled**; contacts are re-tiled inside the new Activ at 0.36 µm pitch (`Cnt.b1` array spacing), grouped by polygon.
- **ThickGateOx** is drawn on the cell boundary grown 0.27 µm in x and 0.42 µm in y — the y-margin proven by the N-row edge experiment (the TGO.a count stays 2 for any N and moves with the outer edge).
- **Five Metal1 re-routes** (`M1E_EDITS` in the generator) resolve the `M1.e` wide-metal gaps the retarget manufactures; each edit is verified in code (polygon count, `M1.a/b/c1`, 5 nm grid).
- **Netlist ↔ layout sync**: the drawn width is authoritative; `work/sync_netlist_widths.py` brings SPICE and CDL to the drawn geometry per *finger* and recomputes parasitics.
- **Site and tracks**: `TRACK_PAD` stretches the mid-cell dead zone to make exactly 17 horizontal 0.42 µm tracks; `pad_to_site` pads each width to the 0.48 µm site (13.75 µm total across the library). Mean cell area is **2.87×** the thin-oxide library (median 2.83, range 1.89–4.21).

4.1 The 18 cells the library-wide map could not produce

`layout_retarget.py` skips 18 cells, and for a long time they were the library's headline limitation. They are not a single hard case: the skips have **one shared root cause**. A p+ source finger butts into the VDD rail tap, so the library-wide y-map cannot scale that band without moving the rail off the cell boundary and breaking abutment. Several flip-flops add a second obstruction, an NMOS Activ band that reaches the library channel cut.

Once that was understood the fix is mechanical, and is applied per cell rather than library-wide:

1. **Cut the butting finger before the map** — the neck joining the p+ source to the rail tap is removed, asserted exactly, so the band is free to scale.
2. **Split the y-insert per cell** — instead of the library's single 0.975 µm insert, each cell distributes the same total across a rail insert, a channel-cut insert and a pSD insert, chosen so that every template band (rails, taps, pSD, N-well, ThickGateOx) lands exactly on its shipped position and the high NMOS band tops out 0.62 µm below the strict N-well bottom.
3. **Restore the finger after the map** at y = 6.99 µm, the butted-junction convention the drawn `slgcp_1` already uses and which the LVS deck connects through `psd_ntap_abutt`.
4. **Rebuild the N-well** with the strict-rule construction of section 6, and re-tile rail contacts onto the site-centred grid.

`work/flop_pilot/gen_seq.py` drives the eight flip-flops from a per-cell insert table; `work/cell_dev/{latches,ebufn,misc}/gen_*.py` cover the four latches, the two tri-state drives, `lgcp_1` and `sighold`. Two cells are not retargets at all: `lgcp_1` is derived from the drawn `slgcp_1` by deleting the scan leg (which also required splitting a shared series node — the scan cell joins the PMOS and NMOS chains in Metal1 where the plain gate does not), and `sighold` is built from scratch on the tie-cell frame. **No device width, length or connectivity deviates from the CDL in any of the 18 cells**, so the existing characterization stays valid; only `area` changes, and it changes from an estimate to a measurement.

Every generated cell had to pass `work/cell_verify.py` before it was merged into the library: LVS against the CDL, both KLayout decks in an abutted mirrored context, Magic, the structural conventions (row height, site-multiple width, N-well rules) and the pin/track report. The gate was itself validated by running it against an already-shipped cell.

5 Functional verification

Suite	Coverage	Result
`verify_logic.py`	60 combinational cells, 452 vectors, both libraries in one deck	**PASS**
`verify_seq.py`	16 stateful cells, 400 clocked samples	**PASS**
`verify_sch.py`	84 schematics + CDL vs SPICE, 920 devices field-by-field	**PASS**

The thin-oxide library is the golden reference — the transform changed devices, never logic. The 12 high-impedance states of the tri-state cells are exercised; illegal set/reset combinations are excluded rather than counted as passes; both suites were re-run on the final, shipped netlists.

6 Physical sign-off

Methodology. Cells are checked in abutted context, never standalone. Two harnesses: `work/make_drc_top.py` (every cell in its own column, mirrored second row at the padded bbox pitch) for cell-internal rules, and `work/make_shared_rail_rows.py` — rows at the true 7.14 µm pitch, orientations N/S/FS, rotated cell order per row, cells advanced by LEF width — for everything only a placed block exhibits. Measurement rules, each bought with a wrong conclusion: one fixed runner invocation (counts from different flag sets are not comparable), diagnosis from flat `klayout.db` rule replication (deep-mode marker coordinates mis-attribute cells), and the IHP thin-oxide library run through the identical harness as control. Pass/fail is parsed from the report database, never the exit code.

DRC. 781 raw violations after the first retarget were driven to zero cell-rule items (density-only residue, same class as the control). One defect class survived the original harness because it could not express it: the retarget left each cell's **rail-tap contacts** at cell-specific x, and in a placed block — where *different* cells share every rail — partially overlapping taps merged into 0.19–0.32 µm bars: ~19 000 `Cnt/CntB` markers on the first `spi_slave` signoff. `work/fix_rail_contacts.py` re-tiles every rail tap onto the site-centred 0.48 µm grid, which both the FS row flip and placement's x-mirror preserve, with in-code guards for implant polarity, enclosures, gate clearance, contact spacing and tap continuity (1 134 → 1 810 rail contacts). Full analysis with figures: the companion report *The Shared-Rail Contact Clash in sg13g2_stdcell_hv* (`shared_rail_contact_fix.pdf`, distributed alongside the library rather than inside it).

N-wells: two rule sets, one layout. The retarget placed the N-well for the *digital* rule set — `NW.c1.dig`, 0.31 µm enclosure of HV p-active, which the IHP rulebook grants inside a `DigiBnd` marker and which the KLayout maximal deck verifies clean. Magic's SG13G2 tech has no DigiBnd concept at all and applies the *analog* rules unconditionally, so it flagged every cell: 1 128 `NW.c1` plus 9 `NW.e1` on a placed counter. Measured against the strict rule the library was short by exactly **25 nm in 65 of 68 cells** (210 nm in `mux4_1`, 215 nm in `slgcp_1`, both of which reach further down).

That margin is recoverable in the well layer alone, so rather than wait on the tool, `work/fix_well_nwc1.py` rebuilds only the N-well: bottom edge 2.625 → 2.570 µm library-wide, a halo around each cell's PMOS active that forces the two deeper jogs automatically, minus halos around NMOS active and p-taps for `NW.d1`/`NW.f1`, and the lateral overhang widened 0.24 → 0.62 µm so the outermost cell of a row is enclosure-clean without a neighbour's merged well (that last point also removed the block-edge `NW.e1` flags). Every rule is asserted per cell in the generator with corner-inclusive square sizing, the worst case of the euclidian deck rules.

Nothing about the devices changes. The well is not a device terminal, characterization runs on the schematic netlist, and the HV PSP model cards carry no well-proximity parameters — so no re-characterization was needed, and only the GDS differs. The result: Magic reports **0 `NW.*`** on the abutted array and both KLayout decks stay clean, i.e. the layout now satisfies the strict and the relaxed interpretation simultaneously. The residual Magic flags (`Cnt.c` contact-overlap, wide-metal `M1.e`) are tool-interpretation differences on geometry both KLayout decks accept. The tool gap itself was reported upstream as IHP-Open-PDK issues #1106 (the modular KLayout deck implements neither `NW.c1` nor `NW.c1.dig`, so its "0 errors" was silent on this rule) and #1107 (Magic lacks the DigiBnd relaxation).

LVS. All 84 cells match, re-run in full after the re-tiling. The four device-less `fill_*` cells need the runner's own `--ignore_top_ports_mismatch` relaxation — IHP's own fill cells fail the identical flow the same way. LVS caught five defect classes dimensional checks could not (merged channels, collapsed notch devices, mis-paired fingers, a dropped substrate tie, stale diode geometry), which is why the per-cell sweep runs after every layout change.

Block-level validation. The `spi_slave` IP, in two variants, was synthesised, placed and routed with this library through LibreLane/OpenROAD at 100 MHz on ~354 × 400 µm dies. Both close timing with zero setup/hold violations at the characterised corner, pass routing DRC, antenna and Netgen LVS in the flow, pass gate-level simulation of the routed netlist (all four SPI modes on the all-modes variant), and pass the **full IHP signoff deck** with zero geometry violations — only the 8 chip-level density/filler markers a filled die resolves at tapeout. The flow settings this takes (flip-flop mapping onto `sdfbbp_1`, excluded cells, Metal1 routing, rails-only fillers) are documented in the `spi-slave-ihp` repository.

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7 Liberty characterisation

`lib/sg13g2_stdcell_hv_typ_3p30V_25C.lib` is produced with **CharLib** against the shipped, layout-synchronised SPICE netlist on the PSP103/OSDI models — 25 839 combinational simulations plus a 2 962-task sequential run: 600 delay/slew tables over 66 cells (52 combinational, 9 flip-flops, 5 latches), none empty. Grids are the thin-oxide grids rescaled by the measured 2.66× delay / 2.20× capacitance ratios; boolean functions are translated from the thin-oxide Liberty and checked by truth-table equivalence. Stock CharLib needed five committed tool workarounds (OSDI shim, `ngspice-shared` backend, case-insensitive supply lookup, Liberty syntax post-pass) and — for sequential cells, which stock CharLib cannot characterise at all — custom clk→Q and setup/hold-bisection procedures with a c2q-degradation pass criterion (`work/seq_delay_procedure.py`).

The shipped file is verified **as data** by `work/verify_lib.py`: structure (600/600 tables populated), cross-view (all pins exist in the CDL), physical (every `area` equals the drawn boundary to 1 nm²), monotonicity along the load axis (1 931/1 932 series, one documented pessimistic waiver: `xnor2_1` glitch latching), input capacitance against an independent both-rails measurement (6.44 vs 5.87 fF, 9.7 % — this check caught a 6.8–7.5× Miller defect in

CharLib's default procedure), and sequential arcs (14/14 cells: delay + setup + hold present, nothing pinned at search bounds). A hand-written transient at a table corner agrees to 0.3 %.

An independent characteriser, **lctime**, was run over eight cells on the same grids and models: in the region STA exercises (real loads, slews < 1 ns) the two agree to a median 2.9 % on delays and 0.0 % on transitions; every disagreement region traces to a stimulus convention, and a direct measurement sides with the shipped table (+0.15 % vs −19 % at the probe point). lctime's input capacitance (+52 %) also corroborates validating Cin against a direct measurement, not a characteriser.

`tiehi`/`tielo` are present in the Liberty without timing arcs (a constant output has none) and with directly measured leakage — `work/tie_leakage.py` applies the sequential cells' settled-tail average to the tie netlists (0.011 / 0.013 nW), so no number is borrowed from `inv_1`.

7.1 Drive limits, and why their absence is not benign

CharLib emits no `max_capacitance` or `max_transition` at all, and the first three revisions of this library shipped without them. That is worse than a missing convenience. OpenSTA answers "no limit" for every pin, so a flow's max-cap and max-slew checks pass **vacuously** — reporting zero violations because there is nothing to violate — and OpenROAD's TritonCTS dereferences an empty buffer-selection result and **segfaults** during clock-tree synthesis, which is how the gap surfaced.

`work/finalize_lib.py` derives the limits from the characterization itself rather than copying the thin-oxide numbers: a pin may not be asked to drive more load than its tables cover, nor to accept a slower edge than was characterized. Per output pin `max_capacitance` is the top of its load axis (0.66–10.56 pF by drive class), per pin `max_transition` the top of the slew axis it was characterized against (6.67 ns combinational, 3.36 ns sequential), with library defaults following the thin-oxide convention of weakest-drive / slowest-edge. The crash is reproducible on demand and was reported upstream with a minimal two-flop test case (OpenROAD issue #11165): a liberty without limits should produce a diagnosable error, not a signal 11.

7.2 The cells CharLib cannot reach

Nine cells are drawn but were never characterized, and the reason is a tool limitation in each case rather than a property of the cell: `gen_charlib_config.py` skips the two statetable clock gates (CharLib has no statetable input form) and the device-less/no-output cells, while the six tri-states are *configured* but silently produce nothing — CharLib has no high-impedance concept anywhere in its schema or code, so it emits an empty result without an error. lctime is no substitute: it can carry a `three_state` attribute but deliberately pins the enable and skips exactly the enable arcs one needs.

They are therefore measured directly against the shipped netlists, the same way the tie and sequential cells already are.

`sighold` (bus holder) is the simplest and is complete: it has no timing arcs at all — the thin-oxide model is `driver_type : bus_hold` plus capacitance and per-state leakage — so it needs leakage (0.0118 nW / 0.0224 nW, ~10^4 below the thin-oxide cell, the same thick-oxide ratio the tie cells show) and a pin capacitance. The capacitance is the interesting part: charge integration, the library's method for ordinary gate pins, is **invalid here**. On a bus holder the integral is dominated by the keeper fighting the driver, so it grows monotonically with the driving edge rate — measured 30.5 fC at a 0.2 ns edge to 86.4 fC at 5 ns, a 2.8× spread — and is a property of the driver, not the cell. A 1 MHz small-signal measurement near the rails gives 0.00367 pF with a 7 % spread across bias; the keeper fight-back charge is recorded separately rather than folded in. The same reasoning explains the thin-oxide cell's 2.8× rise/fall capacitance asymmetry as an artifact of its measurement, so the HV cell ships equal rise and fall values.

The six tri-states are characterized by `work/char_tristate.py` and are in the shipped Liberty. Each carries three arc classes rather than one: the `combinational` data arc, plus `three_state_enable` and `three_state_disable` measured into and out of a floating node defined by a 1 GΩ mid-rail keeper — the construct the functional suite already uses for its 12 Hi-Z checks. The harness is calibrated by re-measuring a shipped cell's pin capacitance through the same code path (0.46 % against the shipped value), and the combinational arcs land inside the library's 2.66× delay band.

The disable arc required a change of measured quantity, not just of threshold. Waiting for a released output to cross a voltage threshold is not physically measurable: once the driver lets go, only the keeper moves the node, so the answer is $0.2\,RC \approx 0.5$ ms and scales with the load. The thin-oxide library cannot have measured it that way either — its disable tables are exactly load-independent, differing by a 1 fs monotonicity epsilon. What is measured here is the drive current going away: from `TE_B` at 50 % to $|I_Z|$ falling through half its on-state value, which is load-independent by construction and needs no synthetic epsilon. Section 7 of the companion characterization report gives the derivation and the ratio tables.

The two clock gates are measured by `work/char_clockgate.py`: a `CLK→GCLK` propagation arc, `setup_rising`/`hold_rising` on the enable pins — the form the sequential bisection procedure already emits — and a `min_pulse_width` constraint on the clock pin, bisected until the gated output loses its pulse, alongside the statetable, internal state pin and `state_function` metadata that make an integrated clock gate usable. *That work is still in progress at the time of writing.* Until it lands the two cells stay excluded from the flow, exactly as the thin-oxide SCL excludes its own clock gates, so nothing downstream can pick a cell it cannot time.

7.3 One deliberate omission: switching power

This library ships **no `internal_power` tables for any cell** — the characterization was scoped to timing and leakage. The special-cell work above kept that scope rather than adding power for nine cells alone: a library where only the tri-states and clock gates carry switching power, and every other cell reports zero, produces power analysis that is worse than uniformly absent because it looks complete. It is recorded as a library-level limitation instead.

8 LibreLane integration

A library that a digital flow cannot load is not usable, and until rev. 5 this one could not be loaded: LibreLane rejects any `STD_CELL_LIBRARY` without a `libs.tech/librelane/<name>/` directory, and the PDK-level `config.tcl` hardcodes the thin-oxide liberty **filenames** in its `LIB` dict while validating every path eagerly. The library now ships the missing pieces.

The SCL (`librelane/`, installed as `libs.tech/librelane/sg13g2_stdcell_hv/`): `config.tcl` with the `CoreSiteHV` site (0.48 × 7.14) and the HV driving, tie, fill, decap, diode and CTS cells; `tracks.info`; latch, mux and tri-state techmaps; and both exclude lists, which mirror the thin-oxide ones. `PDN_RAIL_WIDTH` stays 0.44 µm — the VSS rail is 0.44 µm symmetric about the row edge and the VDD pin shape, though taller, fully covers a 0.44 µm strap.

The PDK config gets a conditional block for `STD_CELL_LIBRARY == sg13g2_stdcell_hv` (single `*_typ_3p30V_25C` corner, `VDD_PIN_VOLTAGE` 3.30), and the library ships a copy of the shared `sg13g2_tech.lef` because that config globs it per-SCL and a Tcl `glob` errors rather than returning empty. Both patches are idempotent.

Flip-flops now map natively. Before all 84 cells were drawn, the only flop with both a liberty and a layout view was the scan cell `sdfbbp_1` — which `dfflibmap` cannot target, because its `next_state`, `(SCE & SCD) | (!SCE & D)`, is not a plain D function. The interim solution was `sdfbbp_map.v`, a techmap covering every posedge Yosys flip-flop type on the scan cell, with the scan mux recycled as the enable mux for `$_DFFE_*` and sync resets folded into the D leg; all 23 clocked mappings were proven equivalent to the Yosys cell semantics with `equiv_induct`. With the `dfrbp`/`dfrbpq`/`sdfrbp`/`sdfrbpq` layouts drawn it is no longer needed — `dfflibmap` maps directly, as in the thin-oxide flow — and it remains only as a documented opt-in for all-scan designs. Removing it cut sequential area on the reference counter by **31 %** (1 699.89 → 1 178.96 µm²), the tie-off overhead the workaround carried.

One lesson worth recording, because it cost two debugging rounds: a techmap applied *after* the synthesis pass's own lowering must write its glue as fine-grained gates (`$_NOT_`, `$_AND_`, `$_OR_`). Inline expressions such as `~E` create coarse cells at a point where no later pass will lower them, and they reach the netlist unmapped. The thin-oxide latch map has the same latent issue; both HV maps avoid it.

A view can be present and still be missing. The spice view was absent from the first three revisions of the upstream PR even though the generator wrote it and the install script copied it every time: the PDK repo's root `.gitignore` carries `*.spice`, and its `libs.ref/.gitignore` re-includes only the `spice/` *directory*, not the file inside it, so `git add` skipped it silently. The thin-oxide library's own spice view predates that rule and is already tracked, so nothing complained — and the failure reproduced only for someone cloning the branch, never for anyone testing an installed tree, which is why every local gate passed. `make_pdk_pr.py` now asks `git check-ignore --no-index` about every installed file and fails the run if any would be invisible; the check reproduces the defect against the unpatched `.gitignore` and passes 191 files against the fixed one.

Tri-states are wired too, now that they carry timing: `SYNTH_TRISTATE_MAP` maps Yosys' `$_TBUF_` onto `sg13g2_hv_ebufn_2` and `TRISTATE_CELLS` lists both footprints, mirroring the thin-oxide SCL.

Block validation. The reference 8-bit counter from the IIC-JKU SG13G2 AMS chip template hardens end to end on the shipped library with no design-level liberty or corner overrides: flops mapped natively, zero unmapped cells, CTS clean, max-cap/max-slew checks non-vacuous and passing, LVS clean through the auto-derived `CELL_SPICE_MODELS`, and **both** KLayout and Magic DRC at zero.

A second design exercises the tri-states specifically — two registered sources driving one shared internal bus, so synthesis must infer `$_TBUF_` and STA must time the enable arcs. It hardens equally clean: 16 `sg13g2_hv_ebufn_2` instances, zero unmapped cells, Magic DRC 0, KLayout DRC 0, LVS 0, routing DRC 0, antenna 0, and zero max-slew and max-cap violations against real limits.

9 Verification summary

Check	Method	Scope	Result
Parasitic formulas	recompute vendor `as/ad/ps/pd`	all 924 thin-oxide devices	PASS, worst error 3.75×10^{-4}
Combinational logic	ngspice vs thin-oxide golden	60 cells, 452 vectors, 12 high-Z states	**PASS**
Sequential logic	ngspice, clocked walk from reset	16 stateful cells, 400 samples	**PASS**
Three-view consistency	symbols vs SPICE vs CDL	84 cells, 920 devices	**PASS**
DRC, padded array	PDK deck, fixed invocation	**84 cells**	**0 cell rules**; 7 density items
DRC, shared-rail array	same deck, true-pitch mixed/mirrored rows	**84 cells** × 4 rows	**0 cell rules**; density only
DRC control	identical harness on IHP `sg13g2_stdcell`	84 cells	0 cell rules, 6 density
DRC, Magic strict analog rules	`drc(full)`, euclidean, abutted array	**84 cells**	**0 `NW.*`**; `Cnt.c/M1.e` tool-interpretation only
LVS	PDK deck, per cell, after the well rebuild	**84 cells** vs CDL	**84/84**
LEF	klayout parse-back + pin sets vs CDL	**84 macros**	PASS, on-grid
P&R block validation	LibreLane `spi_slave` ×2 + full signoff deck	~354 × 400 µm, 100 MHz	

Check	Method	Scope	Result
			all clean; signoff **0 geometry violations**, 8 density markers
Liberty structure/ views/areas	`verify_lib.py` 1–3	76 cells	PASS, **all areas measured**, exact to 1 nm²
Liberty monotonicity	`verify_lib.py` 4	1 932 delay series	1 931 + 1 documented waiver
Liberty Cin	vs both-rails reference	`inv_1`	6.44 vs 5.87 fF (9.7 %)
Liberty delay point-check	hand-written transient	table corner	0.3 %
Independent characteriser	lctime, 8 cells, 3 132 points	STA region	median 2.9 % / 0.0 %
Sequential arcs	`verify_lib.py` 6	14 flip-flops/latches	**14/14 clean**
Site geometry	boundary scan	all **84 cells**	7.140 µm, widths on 0.48 µm site
Tie-cell leakage	`tie_leakage.py` settled-tail average	`tiehi`/`tielo`	0.011 / 0.013 nW, in the Liberty
Pin track alignment	`grid_align_pins.py --apply` + full re-signoff	12 of 25 off-track pins widened	DRC 0 cell rules ×2, LVS 68/68
PDK dev-branch re-run	IHP-Open-PDK dev decks (KLayout 0.30.9)	both DRC arrays, 84-cell LVS	main tables **clean**, LVS **84**/**84**
New-cell signoff gate	`cell_verify.py` per candidate before merge	18 generated cells	**18/18 PASS** (LVS + both decks + Magic + structure + pins)
Liberty drive limits	`verify_lib.py` 7	73 output pins	present on every pin; TritonCTS crash resolved
Liberty special classes	`verify_lib.py` 8	tri-state / ICG / bus-hold constructs	complete-construct check
LibreLane block run	reference counter, no design-level overrides	RTL→GDS	flops native, DRC **0**/**0**, LVS clean
LibreLane tri-state run	shared-bus design inferring `$_TBUF_`	RTL→GDS	16 `ebufn_2`, DRC **0**/**0**, LVS 0, slew/cap 0
Git visibility of views	`git check-ignore --no-index` in `make_pdk_pr.py`	191 installed files	all trackable

The library is submitted upstream as IHP-Open-PDK PR #1103 (draft): `libs.ref/sg13g2_stdcell_hv` following the dev-branch layout, a two-line `xschemrc` registration, and an optional KLayout autorun macro, with the cell-set choice, the macro, and xschem view-machinery unification flagged as maintainer questions. `work/make_pdk_pr.py` assembles and verifies the contribution against a checkout.

10 Known limitations

- **No switching power.** The library ships `leakage_power` but no `internal_power` for any cell (section 7.5). Dynamic power analysis will see only leakage.
- **One corner** (typical, 3.3 V, 25 °C); sequential leakage is a single-settled-state number; the delay cells' ratios shift with the 450 nm minimum; the decaps store less per unit area.
- **Timing is schematic-characterized.** Device sizes match the drawn layout exactly (LVS-proven per cell), but no parasitic extraction is folded back into the tables; a PEX-based re-characterization is the natural next step.
- **11 signal pins remain off the vertical Metal1 track grid** and cannot be widened within `M1.b` (12 others were). Route with `RT_MIN_LAYER` Metal1, or expect the router to jog.
- **Not silicon-proven.** Everything here is simulation against the PDK models. The library has now been independently re-tested by a third party — RTL-to-GDS on the upstream PR, reproducing the DRC, LVS, STA and mapping results — but it has not been fabricated.

11 Reproducing the library

```
cd /foss/designs/sg13g2_stdcell_hv/work

python3 gen_hv_lib.py            # netlists, schematics, symbols, Verilog
python3 gen_gallery.py           # gallery sheet
python3 layout_retarget.py       # gds/ (66 cells; prints the 18 skips)
python3 gen_tie_cells.py         # + tiehi / tielo
python3 flop_pilot/gen_seq.py    # the 8 flip-flops (per-cell y-map recipe)
# latches, ebufn drives, lgcp_1, sighold: cell_dev/*/gen_*.py
python3 cell_verify.py <cell>.gds  # gate every generated cell before merge
python3 fix_rail_contacts.py     # rail taps onto the site-centred grid
python3 sync_netlist_widths.py   # SPICE + CDL follow the drawn geometry
python3 fix_well_nwc1.py         # N-wells to the strict analog HV rules
python3 gen_lef.py               # lef/ (CoreSiteHV + 84 macros)
python3 make_drc_top.py          # padded two-row DRC context
python3 make_shared_rail_rows.py # shared-rail mixed-row DRC context

python3 verify_logic.py; python3 verify_seq.py; python3 verify_sch.py
./run_lvs.sh                     # per-cell LVS, 84/84
```

```
# DRC: PDK run_drc.py on drc/drc_top.gds and drc/shared_rail.gds
#      plus magic drc(full) on drc_top.gds for the strict N-well rules

./run_charlib.sh ../lib/sg13g2_stdcell_hv_typ_3p30V_25C.lib
# sequential cells via charlib_patched.py, then:
python3 merge_lib.py ../lib/sg13g2_stdcell_hv_typ_3p30V_25C.lib <seq.lib>
python3 seq_leakage.py; python3 tie_leakage.py
python3 char_sighold.py         # bus holder: leakage + small-signal Cin
python3 char_tristate.py        # tri-states: data + enable/disable arcs
python3 char_clockgate.py       # ICGs: CLK->GCLK, setup/hold, min pulse
python3 finalize_lib.py         # drive limits, physical stubs, corner name
python3 update_lib_area.py      # areas from the LEF
python3 verify_lib.py           # gates the shipped Liberty as data
python3 lctime_compare.py       # independent characteriser cross-check

python3 make_pdk_pr.py --pdk <IHP-Open-PDK checkout>   # assemble the PR
```

Every number in this report is reproducible from these scripts; the `work/` directory and the `README.md` are the provenance of record.